

DNS Query Tools and SMB Enumeration

Lab Steps

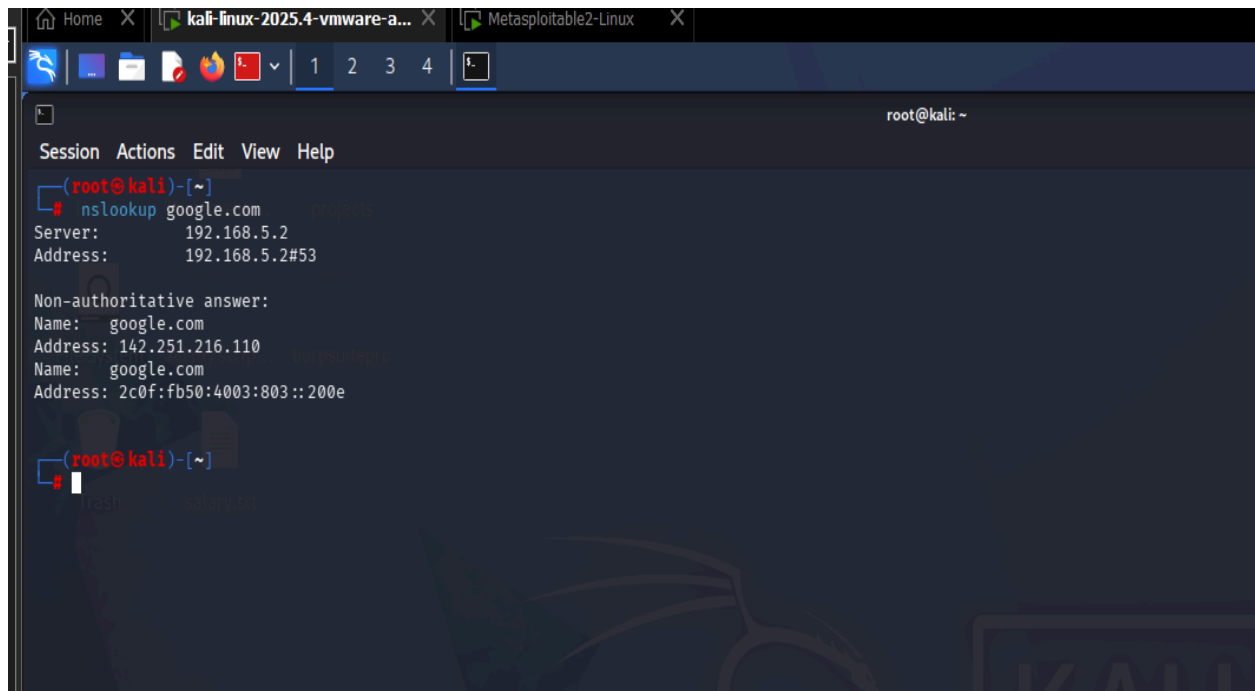
Step 1: DNS Queries with nslookup, host, and dig

1. Using nslookup:

- Open your terminal in Kali Linux.
- Perform a DNS lookup for a live domain:
- Command Used:
 bash
 nslookup google.com

Exercise 1:

- What information did you obtain from the `nslookup` command?
Document the IP addresses and any additional records retrieved.

A screenshot of a Kali Linux terminal window. The window has a dark blue background with a terminal icon in the top left corner. The title bar shows three tabs: 'Home', 'kali-linux-2025.4-vmware-a...', and 'Metasploitable2-Linux'. The terminal window has a menu bar with 'Session', 'Actions', 'Edit', 'View', and 'Help'. The prompt is 'root@kali: ~'. The command 'nslookup google.com' has been entered, and the output is displayed. The output shows two sets of information: one for the authoritative server (192.168.5.2) and one for a non-authoritative answer (142.251.216.110).

```
root@kali: ~  
# nslookup google.com  
Server:      192.168.5.2  
Address:     192.168.5.2#53  
  
Non-authoritative answer:  
Name:   google.com  
Address: 142.251.216.110  
Name:   google.com  
Address: 2c0f:fb50:4003:803::200e
```

Information Obtained:

Server: 192.168.5.2
Address: 192.168.5.2#53

Non-authoritative answer:
Name: google.com
Address: 142.251.216.110
Name: google.com
Address: 2c0f:fb50:4003:803::200e

2. Using host:

- Command Used:
bash
host google.com

Output host

```
(root@kali)-[~]
# host google.com
google.com has address 142.251.216.110
google.com has IPv6 address 2c0f:fb50:4003:803::200e
google.com mail is handled by 10 smtp.google.com.
google.com has HTTP service bindings 1 . alpn="h2,h3"

(root@kali)-[~]
#
```

Information Obtained

- Shows multiple DNS record types in one output:

IPv4 address

IPv6 address

Mail server (MX record)

HTTP service bindings

Exercise 2:

- Compare the output of `host` with `nslookup`. What differences did you observe?

1. host gives more detailed DNS information including MX and service records
2. nslookup mainly returns basic IP address lookup results

3. Using dig

- Perform a more detailed query using `dig`:

bash

dig <target-domain>

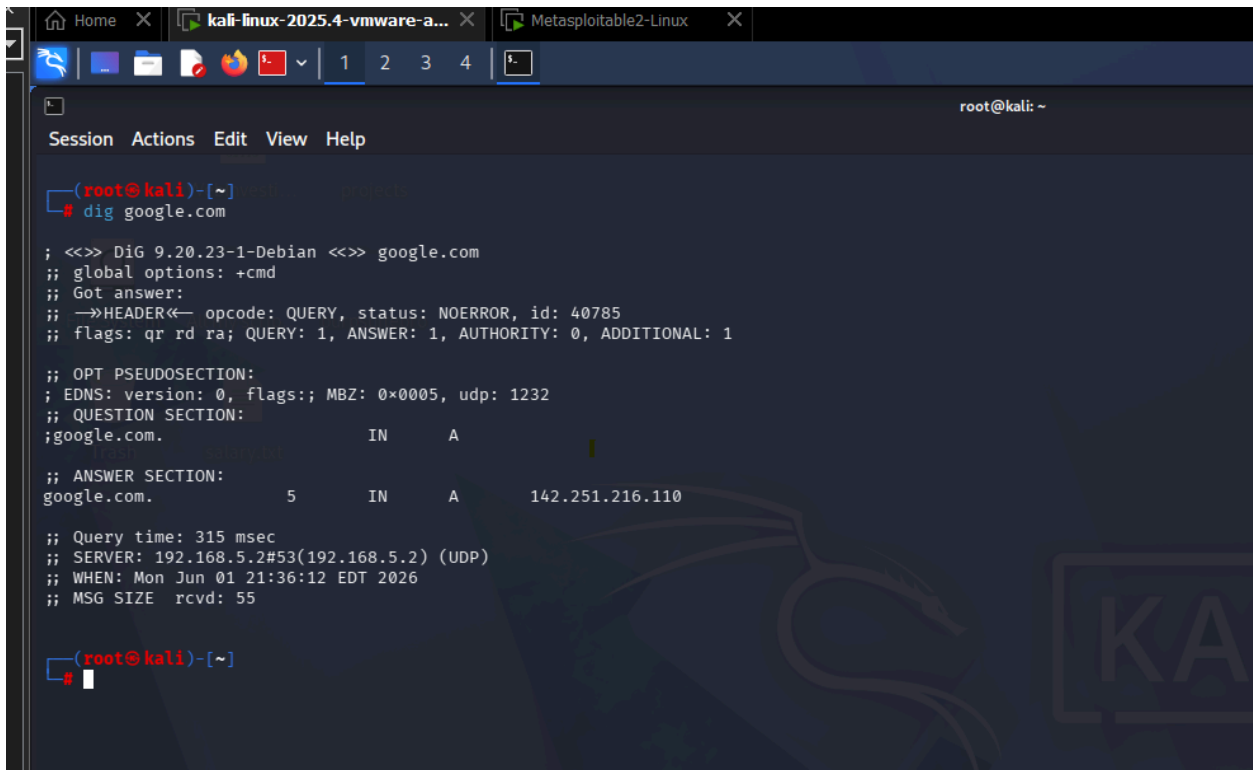
- Command Used:

bash

dig google.com

Exercise 3:

- Analyze the output of the `dig` command. What additional information can you extract compared to the previous tools?



```
(root@kali)-[~]
# dig google.com

; <<>> DiG 9.20.23-1-Debian <<>> google.com
;; global options: +cmd
;; Got answer:
;; -->HEADER<-- opcode: QUERY, status: NOERROR, id: 40785
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; MBZ: 0x0005, udp: 1232
;; QUESTION SECTION:
;google.com.                IN      A
;

;; ANSWER SECTION:
google.com.                 5       IN      A      142.251.216.110

;; Query time: 315 msec
;; SERVER: 192.168.5.2#53(192.168.5.2) (UDP)
;; WHEN: Mon Jun 01 21:36:12 EDT 2026
;; MSG SIZE rcvd: 55

(root@kali)-[~]
#
```

Additional information from dig:

1. The exact DNS query made:google.com. IN A
2. Header info like:opcodestatus: NOERROR ,flags
3. TTL (Time To Live)
4. Shows how long the record is cached (5 seconds here)
5. DNS server used
6. Query time: 315 msec
7. MSG SIZE rcvd: 55
8. Udp used for the query

4. **Advanced DNS Queries:**

- Query specific DNS record types (e.g., MX, TXT):

bash

dig <target-domain> MX

dig <target-domain> TXT

Commands Used

bash

dig 142.251.216.110 MX

dig 142.251.216.110 TXT

```
root@kali: ~  
Session Actions Edit View Help  
root@kali)~# dig 142.251.216.110 MX  
  
;<>> DiG 9.20.23-1-Debian <>> 142.251.216.110 MX  
;; global options: +cmd  
;; Got answer:  
;; ->HEADER<- opcode: QUERY, status: NXDOMAIN, id: 53762  
;; flags: qr rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 1, ADDITIONAL: 1  
  
;; OPT PSEUDOSECTION:  
; EDNS: version: 0, flags:; MBZ: 0x0005, udp: 1280  
;; QUESTION SECTION:  
;142.251.216.110. IN MX  
  
;; AUTHORITY SECTION:  
. 5 IN SOA a.root-servers.net. nstld.verisign-grs.com. 2026060102 1800 900 604800 86400  
  
;; Query time: 426 msec  
;; SERVER: 192.168.5.2#53(192.168.5.2) (UDP)  
;; WHEN: Mon Jun 01 22:17:25 EDT 2026  
;; MSG SIZE rcvd: 119  
  
root@kali)~# dig 142.251.216.110 TXT  
  
;<>> DiG 9.20.23-1-Debian <>> 142.251.216.110 TXT  
;; global options: +cmd  
;; Got answer:  
;; ->HEADER<- opcode: QUERY, status: NXDOMAIN, id: 29735  
;; flags: qr rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 1, ADDITIONAL: 1  
  
;; OPT PSEUDOSECTION:  
; EDNS: version: 0, flags:; MBZ: 0x0005, udp: 1280  
;; QUESTION SECTION:  
;142.251.216.110. IN TXT  
  
;; AUTHORITY SECTION:  
. 5 IN SOA a.root-servers.net. nstld.verisign-grs.com. 2026060102 1800 900 604800 86400  
  
;; Query time: 344 msec  
;; SERVER: 192.168.5.2#53(192.168.5.2) (UDP)  
;; WHEN: Mon Jun 01 22:18:20 EDT 2026  
;; MSG SIZE rcvd: 119  
  
root@kali)~#
```

Information Obtained

dig 142.251.216.110 MX

HEADER - opcode: QUERY, status: NXDOMAIN, id: 53762

flags: qr rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 1,

ADDITIONAL: 1

EDNS: version: 0, flags:; MBZ: 0x0005, udp: 1280

QUESTION SECTION:

142.251.216.110. IN MX

AUTHORITY SECTION:

. 5 IN SOA a.root-servers.net.

nstld.verisign-grs.com. 2026060102 1800 900 604800 86400

Query time: 426 msec

SERVER: 192.168.5.2#53(192.168.5.2) (UDP)

WHEN: Mon Jun 01 22:17:25 EDT 2026
MSG SIZE rcvd: 119

dig 142.251.216.110 TXT

HEADER - opcode: QUERY, status: NXDOMAIN, id: 29735
flags: qr rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 1,
ADDITIONAL: 1
EDNS: version: 0, flags:; MBZ: 0x0005, udp: 1280
QUESTION SECTION:
142.251.216.110. IN TXT
AUTHORITY SECTION:
. 5 IN SOA a.root-servers.net.
nstdld.verisign-grs.com. 2026060102 1800 900 604800 86400
Query time: 344 msec
SERVER: 192.168.5.2#53(192.168.5.2) (UDP)
WHEN: Mon Jun 01 22:18:20 EDT 2026
MSG SIZE rcvd: 119

Exercise 4

- What did you learn from querying different record types? How can this information be useful in a penetration test?

I learned that different DNS record types reveal different information, such as IP addresses, mail servers, and domain services. In a penetration test, this information helps identify potential targets, map network infrastructure, and discover services that may be vulnerable.

Step 2: SMB Enumeration with enum4linux

1. Installing enum4linux (if not already installed):

- Ensure that `enum4linux` is installed on your system:

```
bash
```

```
apt install enum4linux
```

2. **Using enum4linux:

- Perform SMB enumeration on a target IP address:

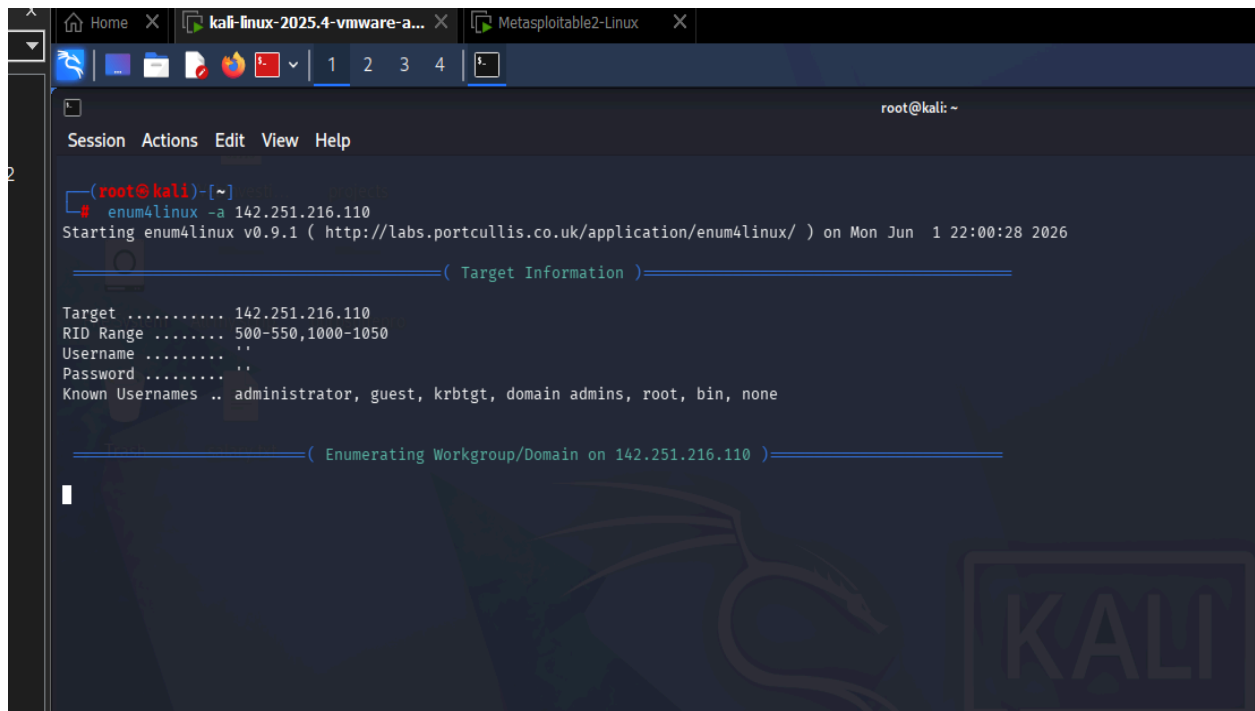
```
bash
```

```
enum4linux -a <target-ip>
```

Command Used:

```
bash
```

```
enum4linux -a 142.252.216.110
```



```
root@kali: ~  
Session Actions Edit View Help  
root@kali)-[~]  
* enum4linux -a 142.251.216.110  
Starting enum4linux v0.9.1 ( http://labs.portcullis.co.uk/application/enum4linux/ ) on Mon Jun 1 22:00:28 2026  
  
===== ( Target Information ) =====  
  
Target ..... 142.251.216.110  
RID Range ..... 500-550,1000-1050  
Username ..... ''  
Password ..... ''  
Known Usernames .. administrator, guest, krbtgt, domain admins, root, bin, none  
  
===== ( Enumerating Workgroup/Domain on 142.251.216.110 ) =====  
  
[KALI]
```

Exercise 5:

- What information did you gather about the target system? Document the shares, users, and any other relevant details found. _____

RID Range - 500-550,1000-1050

Username Not shown

Password

Known Usernames - administrator, guest, krbtgt, domain admins, root, bin,

Showed None.

3. Filtering Results:

- Use specific options to filter results (e.g., listing only shares or users):

bash

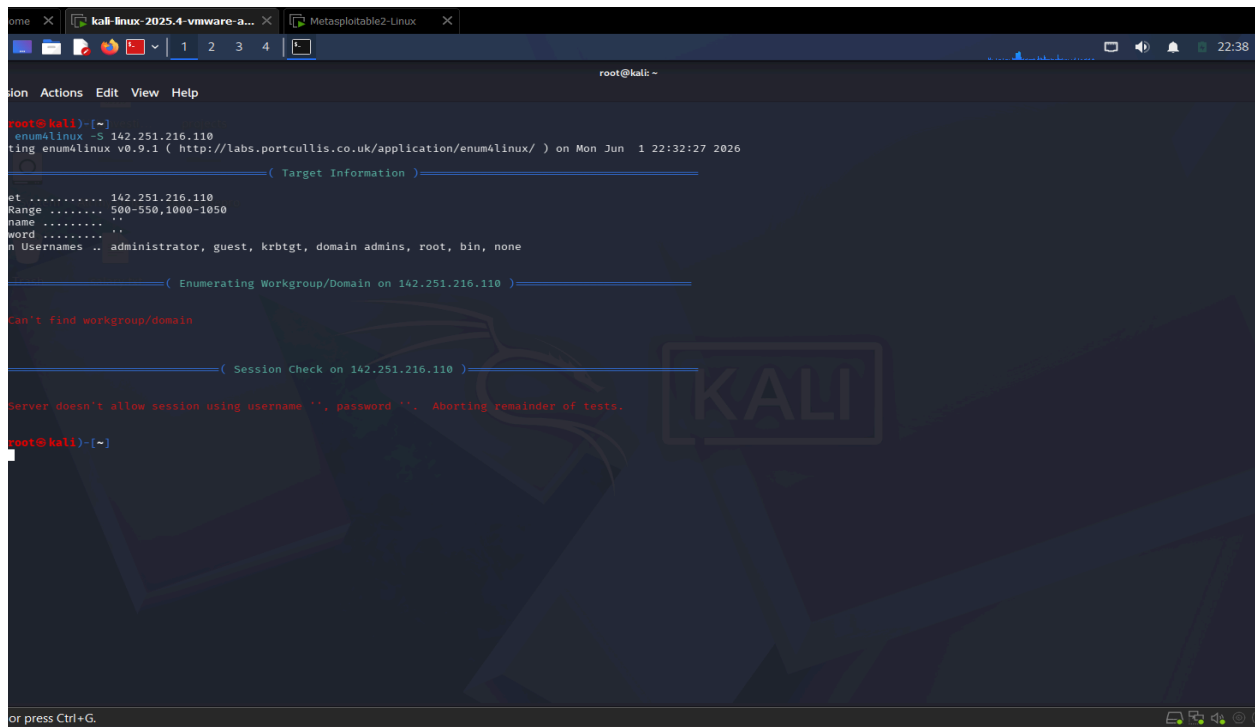
enum4linux -S <target-ip> # Lists shares

enum4linux -u <target-ip> # Lists users

Commands used:

enum4linux -S 142.251.216.110

enum4linux -u 142.251.216.110



```
root@kali: ~  
enum4linux -S 142.251.216.110  
Running enum4linux v0.9.1 ( http://labs.portcullis.co.uk/application/enum4linux/ ) on Mon Jun 1 22:32:27 2026  
  
----- ( Target Information ) -----  
Host ..... 142.251.216.110  
Range ..... 500-550,1000-1050  
Name .....  
Word .....  
Known Usernames .. administrator, guest, krbtgt, domain admins, root, bin, none  
  
----- ( Enumerating Workgroup/Domain on 142.251.216.110 ) -----  
  
Can't find workgroup/domain  
  
----- ( Session Check on 142.251.216.110 ) -----  
  
Server doesn't allow session using username '', password ''. Aborting remainder of tests.  
  
root@kali: ~
```

Displayed

Target 142.251.216.110

RID Range 500-550,1000-1050

Username "

Password "

Known Usernames .. administrator, guest, krbtgt, domain admins,
root, bin, none

Exercise 6:

- Compare the results obtained from `enum4linux` with your findings from DNS queries. What insights can you gain about the target network?

DNS queries displayed relevant details such as :

Server, Address

Non-authoritative answer:

Name: google.com

Address: 142.251.216.110

Name: google.com

Address: 2c0f:fb50:4003:803::200e

While enum4linux didn't show any relevant details

Exercise 7: DNS and SMB Enumeration Report

Methodology Used

Information was gathered through passive and active enumeration techniques. DNS enumeration was performed to identify domain-related records, while SMB enumeration was used to discover shared resources and user information available on the target system.

Tools Employed

- host - Used to retrieve DNS records such as A and MX records.
- nslookup - Used to resolve domain names and verify DNS information.
- dig - Used to obtain detailed DNS query results, including TTL values and query metadata.
- SMB enumeration tools - Used to identify shared folders, users, and accessible network resources.

Findings

DNS Records

- A Record: Resolved the domain to the IPv4 address 142.251.216.110.
- MX Record Query: Returned NXDOMAIN, indicating no MX record exists for the queried IP address.
- TXT Record Query: Also returned NXDOMAIN, indicating no TXT record exists for the queried IP address.

SMB Enumeration

- Identified available SMB shares on the target system.
- Gathered information about accessible resources and user accounts.
- Determined which shares were publicly accessible and which required authentication.

Key Insights

The DNS information provided details about the target's addressing and infrastructure. SMB enumeration revealed network resources that could potentially be accessed by users on the network. Together, these findings help create a clearer picture of the target environment.

Relevance to Penetration Testing

This information is valuable during a penetration test because it helps identify potential attack surfaces, discover exposed services, map network infrastructure, and locate resources that may contain sensitive information or misconfigurations. Proper enumeration enables testers to understand the environment before conducting further security assessments.